



Optimizing Depot Performance

Investigating operational throughput drop during June 25 to July 1, 2026 — flow rates fell significantly across multiple key depots, threatening daily distribution targets.

- **PRESENTED BY**

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01



CSV LOGS VIA PANDAS

Operational CSV logs from each depot were ingested and processed using pandas, enabling structured parsing of flow rate records and timestamps across all monitored sites.

02



WEATHER API HOURLY DATA

Hourly weather data was retrieved via an external Weather API, capturing rainfall, temperature, and humidity readings aligned to each depot's geographic coordinates.

03



SQLITE SHIFT DATABASE

Shift schedules and workforce assignment records were extracted from a SQLite relational database, providing granular visibility into staffing patterns per depot per day.

04



MASTER DATASET ALIGNED

All three independent data sources were merged into a single master dataset, unified by timestamp to enable cross-variable analysis of throughput, weather, and staffing.

Data Integration Architecture

A robust ETL pipeline was developed to consolidate three independent data sources — CSV operational logs, Weather API feeds, and SQLite shift records — into one unified master dataset, enabling precise timestamp-aligned analysis of depot performance during June 25 to July 1, 2026.

Insights & Recommendations



WEATHER IMPACT

Rainfall shows zero correlation with flow rates. Pumps are physically shielded — weather is not a factor.

STAFFING REDUCTION

Night shifts were reduced by half during the affected period. Holiday nights compounded the problem significantly.

KEY RECOMMENDATION

Increase night staff to six operators and add supervisor coverage for all holiday nights.

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RAINFALL CORRELATION
COEFFICIENT

WEATHER DOES NOT AFFECT FLOW

01 Analysis confirms a zero correlation coefficient between rainfall and depot flow rates. All pump infrastructure is physically shielded from weather conditions, ruling out environmental factors as a cause of the June 25 – July 1 throughput drop.

NIGHT SHIFT STAFFING DEFICIT

02 Night shift headcount was reduced by 50% during the affected week. Holiday nights showed the most severe impact on distribution targets, with throughput falling furthest when reduced crews coincided with peak demand periods.

STRATEGIC RECOMMENDATIONS

03 Increase night staff to a minimum of six operators per shift. Deploy dedicated supervisors on all public holiday nights. Implement weekly staffing level monitoring to proactively address shortfalls before they impact daily distribution targets.